Evidence, not verdicts: what 590 public PX4 crash logs can and cannot tell a drone fleet operator

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Abstract

Flight logs are widely expected to determine whether a drone crash was caused by the pilot or the aircraft. Across 590 public PX4 crash logs from 479,547 flights, we evaluate that expectation using declared protocols committed prior to analysis. First, automatic crash attribution between pilot and aircraft is not achievable above chance: a fixed rule reached a balanced accuracy of 0.47 on held-out crashes, cross-validated logistic regression yielded AUCs of 0.45–0.46 (permutation null $p = 0.51$), and gradient-boosted trees remained near chance once impact was excluded. Second, diagnostic checks suffer from crash data leakage: tuning fault detection fell from 77.8% to 11.1% sensitivity ($p = 0.031$) without the final 10 s, and cross-sectional separation was substantially attenuated within the same drone (AUC 0.58). Third, drones that later crashed showed higher baseline vibration than matched peers on older estimator metrics (mean rank 0.67, Holm $p = 0.016$), though replication on newer accelerometer metrics was inconclusive ($p = 0.25$). Flagging upper-quartile vibration identifies 63.3% of crashed airframes (matched-set PPV 1.04% [95% CI 0.83–1.25%] at the 0.74% vehicle base rate; fleet-wide reference PPV 3.8% at the 1.54% log rate, 26 inspections per crash). AeroFleet 2.0, an open-source assurance tool, presents longitudinal evidence without issuing automatic verdicts.

Keywords: Unmanned aircraft systems, Flight log analysis, Crash investigation, Fault diagnosis, Data leakage, Declared protocols, Fleet assurance

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1. Introduction

Small unmanned aircraft are now flown commercially for survey, inspection, agriculture and delivery, and the rules that govern them place weight on records. In the European Union and the United Kingdom an operator in the specific category must keep records of operations, maintenance and personnel for at least three years [1, 2]; India requires an accident to be reported within 48 hours [3], and the United States within ten days [4]. When something goes wrong, the onboard flight log is the richest record available. Open-source log viewers, commercial fleet platforms and, more recently, machine-learning and language-model prototypes promise to turn that log into an answer: what failed, and whether the pilot or the aircraft was responsible [5, 6].

The question matters beyond engineering curiosity. In aviation safety science, accidents are recognized as systemic, multi-causal events involving human factors, environmental conditions and organizational influences [7]. Commercial warranty claims, insurance settlements and regulatory filings nevertheless frequently compress these dynamics into a binary decision: whether the pilot or the aircraft was primarily responsible. An automatic verdict that looks objective can carry weight it has not earned. Fleet operators face a related but different question: not why one flight ended badly, but which of their drones are most likely to fail next and should be inspected first. A log-based method that is good at the second task need not be good at the first, and confusing the two can mislead both investigators and maintainers.

Whether logs can deliver an answer to the first question has, to our knowledge, not been tested on real crashes at scale. Public fault datasets are dominated by simulation or by faults injected on purpose [8, 9, 10], and accident statistics classify outcomes such as loss of control rather than causes visible in a log [11, 12]. Two pitfalls make optimistic results easy to obtain. First, a log that contains a crash also contains its consequences: saturated motors, extreme attitudes and estimator errors. A detector can learn these instead of the cause, a form of data leakage [13, 14]. Second, comparing crash flights with other operators' good flights mixes the flight with the aircraft and the people using it, so a difference between groups need not say anything about the flight that crashed.

This paper asks four questions of the largest public archive of real autopilot logs, PX4 Flight Review [15, 16]:

Q1 Do configuration settings differ between crash flights and normal flights?

Q2 Can a log separate crashes caused by the pilot from crashes caused by the aircraft?

Q3 Does the flight that crashed show warning signs relative to the same drone's history, or are the differences between drones?

Q4 Do expert diagnoses agree with uploader labels, and do automatic checks find the faults experts name?

Because language-model assistants are now proposed for exactly this task, we also report a controlled test of small local models used as a review panel.

The contributions are as follows. (i) We assemble a real-flight crash benchmark from public logs and check its labels against expert diagnoses. (ii) Every confirmatory test was declared, with its threshold and pass mark, in public version control before the data it used were analysed, and decisive tests were repeated on held-out data, following declared test protocols committed prior to analysis [17]; secondary sensitivity and diagnostic analyses added during revision are explicitly designated as exploratory. (iii) We show that crash attribution from logs was not achievable with these labels and features, and that apparent diagnostic accuracy can come from the crash itself. (iv) We report one positive screening signal: drones that later crash are rougher on their ordinary flights than matched drones on older estimator metrics, though replication on newer accelerometer metrics was inconclusive ($p = 0.25$) and fleet screening yields an operational positive predictive value of 1.04% under empirical matched-set screening (relative risk 2.09×, lift 1.40×; 3.8% under a theoretical fleet-wide reference, relative risk 5.02×). (v) We show how these results translate into the design of an open-source assurance tool, AeroFleet 2.0, and report engineering checks of its parts, including one negative result (Section 6). Code, declarations, derived features and labels are released.

2. Related work

Fault datasets. Table 1 compares our data with the main public resources. ALFA provides 47 fixed-wing flights with injected engine and control-surface failures [8]. RflyMAD covers 5,629 multicopter fault cases with eleven fault types, of which 497 are real flights and the rest software- or hardware-in-the-loop simulations [9]. UAV-SEAD offers about 1,396 real PX4 logs annotated for state-estimation anomalies [10], used by later anomaly-mining work [18]; a recent testbed quantifies the discrepancies between simulated and real PX4

Table 1: Public UAV fault and anomaly data compared with this study.

Resource	Flights	Faults or labels	Real crashes labelled by cause
ALFA [8]	47 fixed-wing flights	Injected engine and control-surface failures	No
RflyMAD [9]	5,629 cases (497 real)	11 injected fault types	No
UAV-SEAD [10]	about 1,396 real PX4 logs	State-estimation anomalies (4 classes)	No
This study	479,547-log index; 590 crash-rated logs; 148 expert-diagnosed threads	Uploader responsibility labels; expert causes	Yes (uploader; partly expert-checked)

logs [19]. These resources target anomaly or fault detection under known faults. None labels real crashes by responsibility, and none separates the pre-crash window from the crash.

Fault detection and diagnosis. Surveys of fault detection for UAVs and robots cover model-based and data-driven methods and stress how hard robotic diagnosis is in the field [20, 21]. Causal models built from autopilot documentation have been used to find actual causes of failures in ArduPilot logs [22]. A study of 569 bugs in PX4 and ArduPilot found parameter handling among the root-cause types of drone-specific bugs [23]. Maintenance-oriented reliability analysis of UAVs models the passage from partial degradation to failure [24], the view our between-drone result supports.

Accident analysis. Analyses of civil drone occurrences find equipment problems and loss of control prominent [11]. Human-factors frameworks for remotely piloted aircraft (such as the Human Factors Analysis and Classification System, HFACS) model mishaps as multi-causal sequences involving organizational influences, environmental preconditions, and supervisory failures alongside direct operator actions [7, 25]. In operational practice, post-incident reporting and warranty claims often collapse these complex systemic interactions into a single nominal party: pilot versus aircraft. Because such

classifications depend heavily on the rater and framing, we evaluate agreement against expert community diagnoses.

Language models for diagnosis. Language-model agents have been applied to simulation testing and log analytics for small UAVs [5], and learned models to classifying problems in drone log messages for forensics [6]. Multi-agent debate has been proposed to improve the factuality of model outputs [26], while indirect prompt injection shows that text inside data can steer model behaviour [27]. We test both properties on our task.

Methodology. Leakage, where information unavailable at prediction time enters training or evaluation, is a known source of irreproducible machine-learning results [13, 14]; label noise likewise biases evaluation [28]. Matching reduces confounding in observational comparisons [29], and declaring hypotheses and analysis scripts in version control prior to running tests separates confirmatory from exploratory analysis [17]. We apply these ideas to drone logs.

Tools for operators. Retrieving similar past cases and reusing their solutions is the basis of case-based reasoning [30]; ranking text by word overlap with BM25 is a standard, deterministic retrieval method [31]. A hand-held miniature of the environment lets a person grasp and inspect a spatial scene at a glance [32], the idea behind the tabletop view in Section 6. We use these as engineering choices and test them only as far as stated.

3. Data

3.1. Archive and labels

PX4 Flight Review is a public service to which PX4 users upload onboard logs (ULog format) for plotting and sharing [16]. We downloaded its public index on 20 October 2026: 479,547 logs (32,793,172 bytes, SHA-256 `1bbb01c3…a92d6`), of which 442,929 came from real hardware rather than simulation, from 38,320 distinct vehicle identifiers. Uploaders may rate a log. A total of 590 logs carry an uploader crash rating (402 aircraft-caused, 188 pilot-caused); Table 2 details the 520 crash-rated logs from real hardware with complete duration and firmware version records. These uploader-assigned responsibility labels are not independently verified ground truth; they reflect the uploaders' own assessments. A notable discrepancy appears between uploader ratings and expert conclusions: while uploaders attributed

31.9% of rated crashes to the pilot (188 of 590), forum experts blamed the pilot in only 6.1% of all threads (9 of 148) and 8.6% of decided threads (9 of 105, Section 5.6). On the 14 overlapping logs carrying an uploader aircraft crash rating, experts agreed in 11 and were unclear in 3 (none blamed the pilot). This gap suggests potential politeness bias on public developer forums, selection bias in which incidents operators submit for community assistance, or a narrower definition of pilot error among autopilot developers compared to end users. Furthermore, while expert coding showed strong agreement on aircraft-rated logs, forum threads did not provide an expert-diagnosed pilot-rated sample to verify the validity of pilot-blamed labels independently. Logs rated "good" or "great" served as normal flights. Logs were parsed with pyulog. We store derived features and log identifiers, not the logs.

A ULog file contains the parameters in force when logging started, any parameter changes during logging, timestamped messages published by the autopilot's modules, and logged text messages with a severity level. The analyses below use the vehicle status (arming state, navigation mode, failsafe flags), attitude and body-rate topics, rate setpoints, inertial-sensor status, actuator outputs, battery status, GPS position and the land detector. The vehicle identifier in the index is derived from the flight controller's hardware serial number; it identifies a flight controller, which in practice usually means one airframe, though a controller can be moved between frames.

Crash-rated logs differ from well-rated logs in composition (Table 2): they are shorter (median 54 s against 388 s, partly because a crash ends the log) and spread over more firmware generations. All between-group comparisons below therefore use either matching or within-drone designs, and we do not pool across firmware generations where measurements differ.

3.2. Samples

Figure 1 and Table 3 show how each analysis sample was drawn. We distinguish two distinct cohorts that each happen to number 148: (1) the leak-curve and attribution modeling sample of 148 crashes and 139 normal flights at $m = 0$ s (Section 4), and (2) the expert community sample of 148 crash discussion threads on the PX4 forum (Section 5.6). The attribution rule was evaluated on 205 crashes (93 development and 112 held-out). All 148 crashes entering the statistical attribution models are a strict complete-case subset of these 205 crashes (100% overlap); 57 crashes were excluded from modeling because required ULog topics were unrecorded or the flight window before cut-off was shorter than 10 s. In the leak-curve and attribution

Table 2: Composition of crash-rated and well-rated logs from real hardware in the archive.

	Crash-rated	Rated good or great
Logs	520	21,112
Median duration	54 s	388 s
Most frequent years	2018 (91), 2025 (89), 2019 (87), 2017 (57)	2018 (10,571), 2017 (5,748), 2019 (2,910)
Most frequent PX4 versions	1.8 (70), 1.9 (66), 1.15 (65), 1.11 (65)	1.7 (6,299), 1.8 (5,785), 1.6 (4,265)
Vehicle types	quadrotor 320, fixed wing 71, hexarotor 38	quadrotor 19,890, fixed wing 620, hexarotor 291

modeling sample, sample sizes across cut-off margins $m \in \{0, 2, 5, 10, 20, 30\}$ s reflect progressive attrition as flights with duration under 10 s before cut-off are dropped: for Task A (crash vs. normal), 148 vs. 139 ($m = 0$ s), 142 vs. 137 ($m = 2$ s), 135 vs. 133 ($m = 5$ s), 114 vs. 130 ($m = 10$ s), 100 vs. 128 ($m = 20$ s), and 89 vs. 124 ($m = 30$ s); for Task B (drone vs. pilot responsibility among crashes), 112 vs. 36 ($m = 0$ s, $n = 148$), 106 vs. 36 ($m = 2$ s, $n = 142$), 101 vs. 34 ($m = 5$ s, $n = 135$), 87 vs. 27 ($m = 10$ s, $n = 114$), 73 vs. 27 ($m = 20$ s, $n = 100$), and 63 vs. 26 ($m = 30$ s, $n = 89$). A constant complete-case subset of 213 flights (89 crashes and 124 normal flights; 63 drone and 26 pilot crashes) with pre-crash duration ≥ 40 s was evaluated across all six margins without attrition.

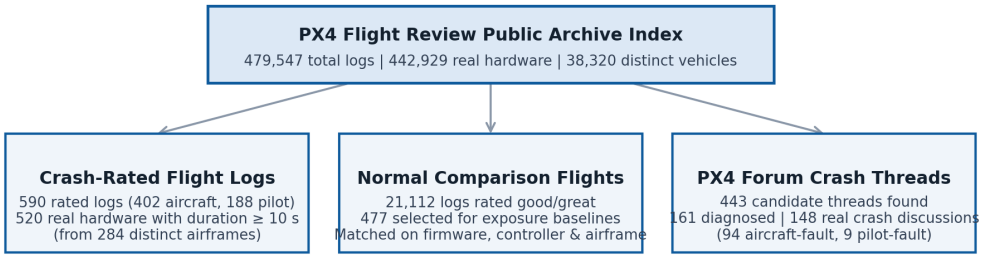
Initial cross-sectional samples for Q1 and Q2 selected flights with PX4 1.10 or newer and at least 5 s of flight; normal flights were matched to crashes on vehicle type and drawn from vehicles without any crash rating. For the longitudinal within-drone and between-drone tests (Q3b, Q3c), only the eligible case count ($n = 12$), not any outcome or feature data, was inspected before deciding to lift the firmware restriction to avoid sample starvation; flights were instead strictly matched within each drone's firmware generation and measurement source. For the within-drone test every crash with a vehicle identifier and at least three earlier non-crash flights of 30 s or more was eligible (101 crashes; up to the five most recent earlier flights each). For the between-drone test each crashed drone was matched with up to two drones without any crash-rated log that shared the flight-controller hardware, the PX4 major.minor version and the vehicle type (66 matched sets, 124 control drones, three randomly chosen flights each, seed 7). Control drones were

sampled without replacement, so no control served two cases. The three exposure flights of a crashed drone were its most recent non-crash flights dated on or before the crash; flights on the crash day were included in the declared test because the archive does not give their order within a day, and removed in a sensitivity analysis.

Not all selected flights enter every test. In the within-drone sample (101 crashes selected), 55 logs could not be read or had under 10 s before the cut-off; a drone enters a feature's test only if the crash flight and earlier flights have that feature from the same measurement source, and rate-tracking error is rotary-wing only and requires rate setpoints, leaving 61 analysed drones for vibration and 46 for tracking. In the between-drone sample (66 matched sets), 71 flights were unreadable or too short; a drone's value requires at least two readable flights, and case and controls must share the measurement source, leaving 60 analysed sets for vibration and 47 for tracking. In the between-drone sample, 124 matched control drones contributed 372 flights (124×3), distinct from the 372 flights (67 crash flights and 305 earlier flights) evaluated in the within-drone predictive model.

3.3. Expert diagnoses

We searched the PX4 discussion forum for threads that link a Flight Review log and mention a crash (443 threads, 410 with a log link). For the 161 threads with a reply from PX4 staff, the development team or a member at the forum's highest trust levels (148 of them describing a real crash), we recorded the expert's conclusion (aircraft, pilot, both, unclear, no diagnosis), the cause named, who gave it, whether the original poster confirmed it, and the sentence stating the diagnosis. Recording was AI-assisted and reviewed by an author; 123 of 124 recorded sentences were found verbatim in their thread by script. A second author then coded all threads blind to the first recording, with the same categories; a script confirmed that 150 of 151 coded rows quoted their own thread or recorded no diagnosis. On the 147 threads both coded, agreement (Cohen's kappa, 95% bootstrap interval) was 0.81 (0.71–0.88) for the expert's conclusion, 0.73 (0.65–0.81) for the cause and 0.79 (0.69–0.88) for whether the poster confirmed it, above the declared pass mark of 0.60. The two coders settled the 45 threads that differed on conclusion or cause by discussion, which changed 13 labels; counts and tests below use the settled labels, and the first recording led to the same conclusions (both are in the repository). The searches combined the Flight Review host names with the words crash, crashed, flyaway, fell, flipped and lost control, and



Analysis Stream (Protocol)	Initial Candidate Pool	Exclusions & Attrition Criteria	Analyzed Sample Size
Q1: Parameter Settings	100 crash, 477 normal	43% crash logs lack factory default recor	57 crash, 113 normal
Q2: Fixed Rule Attribution	205 crashes, 143 normal	None (evaluates all available crashes: 93	205 crashes, 143 normal
Q2/Q3a: Data Leakage Cur	205 crashes, 143 normal	57 dropped (missing topics or duration <	148 crash, 139 normal (m=0 s 89 crash, 124 normal (m=30 s
Q3b: Within-Drone Longitu	101 crashed drones	34 dropped (single flight recorded or dur	67 multi-flight drones (372 flig 61 vibration sets, 46 tracking s
Q3c: Between-Drone Matcl	66 matched sets (101 ca	6 sets dropped (unmatched sensor metri	60 vibration sets (124 controls 47 rate-tracking sets
Q4: Expert Community Cas	148 real crash threads	110 logs unhosted or expired download l	38 downloadable logs (all aircr 30 with complete telemetry fe

Figure 1: From the public index to each analysis. Counts are logs unless stated.

every returned thread was retrieved through the forum's public interface. A reply counted as expert when its author was marked as staff, belonged to the development team group or held one of the forum's two highest trust levels. Only public posts were used; usernames and post text are not redistributed.

4. Methods

4.1. Declared test protocols

For each test, the selection, features, statistic, threshold and pass mark were committed and pushed to a public git repository before the data it used were downloaded or analysed. Unlike third-party registries (such as the Open Science Framework, Zenodo or Software Heritage), git commits on self-hosted repositories remain under author control until an archived DOI release is minted on acceptance; commit identifiers and hashes are preserved in public version control. One amendment was data-dependent: after

observing that the firmware filter of the within-drone test left only 12 eligible cases (without inspecting any outcome data), the firmware filter was removed and a fallback for older firmware was added. One coding error was corrected after a first run (logs whose CPU-load record lacked the load field were being skipped); no threshold changed. Secondary sensitivity analyses and diagnostic evaluations added during peer review are explicitly designated as exploratory and documented with commit hashes in Section 5.8.

4.2. Crash moment and analysis window

For rotary-wing aircraft the crash moment t_c is the first time after take-off at which the tilt $\theta = \arccos\left(1 - 2(q_x^2 + q_y^2)\right)$, computed from the attitude quaternion, exceeds 60°; for fixed-wing aircraft, and when no such tilt occurs, it is the end of the armed period. Normal flights use the end of the armed period. Take-off is the first sample after arming at which the land detector reports the vehicle airborne. An analysis with cut-off m uses the window from take-off to $t_c - m$ and is skipped if the window is shorter than 10 s. Because normal flights include the complete descent, landing and disarm sequence while crash windows terminate at impact or tilt, this introduces an inherent windowing asymmetry that contributes to differences in duration and whole-window aggregates.

4.3. Features

Table 4 defines the features. Vibration is not measured the same way across firmware generations: newer logs report an accelerometer vibration metric, older ones an estimator vibration estimate. Comparisons are therefore only made between flights measured with the same source, and the source is recorded with every value. The rate-tracking error falls back to the attitude message's body rates when the angular-velocity topic is absent.

4.4. Tests

Q1 settings. For logs that carry the firmware's record of default values, a protection was counted as disabled when its value differed from the default and switched it off (for example a failsafe action of "none"). Crash and normal logs were compared with Fisher's exact test.

Q2 attribution. Task B is defined as the binary attribution of crashes between pilot and aircraft fault. The declared rule labelled a crash "aircraft" if any of eight signs fired before t_c (a critical battery warning or remaining charge below 10%; any motor at 95% or more for at least 1 s in the last 10 s; more than 50 clipping events; poor GPS in a GPS-dependent mode; a failure-detector flag; an error-level message; CPU load above 95%; an active failsafe), otherwise "pilot" if the mode at t_c was pilot-flown, otherwise "unknown". The pass mark was a balanced accuracy of at least 0.70 on decided cases, answering at least 60% of crashes and being wrong on at most 20%. A model-based check used L2-regularised logistic regression on all features ($\lambda = 1$, standardised inputs, median imputation within training folds), evaluated by stratified five-fold cross-validation repeated five times and summarised by the ROC AUC [33]. As an exploratory sensitivity analysis, non-linear feature interactions were evaluated using depth-3 gradient-boosted decision trees.

The signs were chosen before the analysis to cover common failure modes of PX4 multirotors: power, propulsion, vibration, navigation, software and failsafes. The pilot rule was deliberately simple: if no aircraft sign fired and the vehicle was in a mode in which the pilot controls attitude or position directly, the crash was attributed to the pilot. The rule was fixed before it was run on any crash log; only the list of message types available in the logs had been inspected.

Q3a crash window. The same model separated crash from normal flights at cut-offs $m \in \{0, 2, 5, 10, 20, 30\}$ s. We declared that the AUC would fall by at least 0.10 between $m = 0$ and $m = 10$ s if crash consequences drove it.

Q3b and Q3c rank tests. For a crashed drone i with value x_{i0} and comparison values $x_{i1}, \ldots, x_{in_i}$, the scaled rank is

$$r_i = \frac{1}{n_i} \sum_{j=1}^{n_i} \mathbf{1}\left[x_{ij} < x_{i0}\right], \tag{1}$$

which is 0 when the crashed drone has the lowest value and 1 when it has the highest; ties count against the hypothesis. A rank statistic was chosen for three reasons: the two vibration sources have different units and scales, so only orderings are comparable across drones; the number of comparisons differs between drones, which the exact null handles directly; and the distributions are skewed, with a few very rough flights, which would dominate a mean difference. In the within-drone test (Q3b) x_{i0} is the crash flight,

cut 10 s before t_c, and the comparisons are the same drone's earlier flights, each cut 10 s before the end of its armed period. In the between-drone test (Q3c) x_{i0} is the median over the crashed drone's three most recent pre-crash flights and the comparisons are the medians of the matched controls. Under no difference each r_i is uniform on $\{0, 1/n_i, \ldots, 1\}$ with mean 0.5, and $\bar{r}$ estimates the probability that the crashed drone exceeds a comparison, the quantity estimated by the Mann–Whitney statistic [34]. One-sided p-values come from 100,000 draws of this exact null; 95% confidence intervals from 10,000 bootstrap resamples of drones [35]. For the proportion of sets in which the crashed drone is the roughest, the expected null rate depends on the mix of set sizes: among the 60 analysed vibration sets, 35 had two matched controls (null probability 1/3) and 25 had one control (null probability 1/2), giving an exact Poisson-binomial expected null proportion of $(35 \times \frac{1}{3} + 25 \times \frac{1}{2})/60 = 40.28\%$ (null mean 24.167, SD 3.745). For duration adjustment in sensitivity analysis, log(vibration) was regressed on log(flight duration) within each measurement source across all exposure flights, and the rank test was repeated on the residuals. Each test had two primary features (vibration, rate-tracking error) at $\alpha = 0.025$.

Q4 expert diagnoses. Expert conclusions were compared with uploader labels. Fixed checks for the fault the expert named (tuning, vibration, GPS or compass, motor, battery) were declared useful if they fired on at least 50% of the logs with that diagnosis and at most 10% of normal flights, first with the full window and then with $m = 10$ s.

Language-model panel. The panel tested a design proposed for advisory diagnosis: several reviewers, each restricted to one slice of evidence (navigation, pilot inputs, GPS, battery, settings), vote on a fixed list of six causes and must cite numbered facts extracted from the log. Robustness was evaluated by replacing one reviewer with an adversary arguing for a chosen wrong cause, and planting hostile strings in uploader-controlled fields. Full details are provided in Appendix B.

Multiplicity. Seven confirmatory tests produce p-values. Holm's procedure [36] is reported over all seven in addition to each test's declared α.

5. Results

5.1. Configuration settings (Q1)

Of 57 crash logs and 113 normal logs carrying a firmware-default record, 29 (51%) and 52 (46%) had at least one protection disabled relative to the default (Fisher $p = 0.63$). Pilot-caused crashes had a protection disabled more often than aircraft-caused ones: 14 of 20 (70%) against 15 of 37 (41%), an observed difference of +29.5% (95% CI 3.9–55.0%, odds ratio 3.42, Fisher's exact two-sided $p = 0.052$). This contrast is exploratory, not part of the pre-declared confirmatory family, and does not reach statistical significance unadjusted (Fisher's exact two-sided $p = 0.052$). Furthermore, default parameter records were present only on newer firmware (missing for 43% of the 100 crash logs); on the 57 available cases, treating disabled protections as a pilot-attribution classifier yields a sensitivity of 70.0%, a specificity of 59.5%, and a balanced accuracy of 0.647, below the declared 0.70 pass mark. None of 100 crash logs changed a safety-critical or flight-limit setting during the flight.

5.2. Attributing a crash to the pilot or the aircraft (Q2)

The fixed rule reached a balanced accuracy of 0.57 on 93 development crashes (35% wrong) and 0.47 on 112 held-out crashes (47% wrong), against the declared 0.70, and labelled 39% of 143 normal flights as aircraft faults (Table 5). The signs fired at similar rates whichever party the uploader blamed (Fig. 2): motor saturation in 23% of aircraft-caused and 29% of pilot-caused crashes (1% of normal flights), battery warnings in 15% and 18% (3%). Error messages and failsafes were no more frequent in crashes than in normal flights. Two signals noticed after the development run did not replicate on held-out crashes: accelerometer clipping (24% against 8%, $p = 0.045$) and manual mode at the crash (18% against 10%, $p = 0.165$).

In Task B, sample sizes decline across margins as flights under 10 s are excluded: 112 drone and 36 pilot crashes at $m = 0\,\mathrm{s}$ ($n = 148$), 106 vs. 36 at $m = 2\,\mathrm{s}$ ($n = 142$), 101 vs. 34 at $m = 5\,\mathrm{s}$ ($n = 135$), 87 vs. 27 at $m = 10\,\mathrm{s}$ ($n = 114$), 73 vs. 27 at $m = 20\,\mathrm{s}$ ($n = 100$), and 63 vs. 26 at $m = 30\,\mathrm{s}$ ($n = 89$). Across these complete-feature cases, the cross-validated logistic regression model on Task B yielded ROC AUCs between 0.449 and 0.458 (0.450 ± 0.015 at $m = 0\,\mathrm{s}$, 0.449 ± 0.022 at $m = 10\,\mathrm{s}$, and 0.458 ± 0.034 at $m = 30\,\mathrm{s}$; Fig. 3). To test whether these below-chance AUCs represent an inverted physical signal or finite-sample noise, we computed a

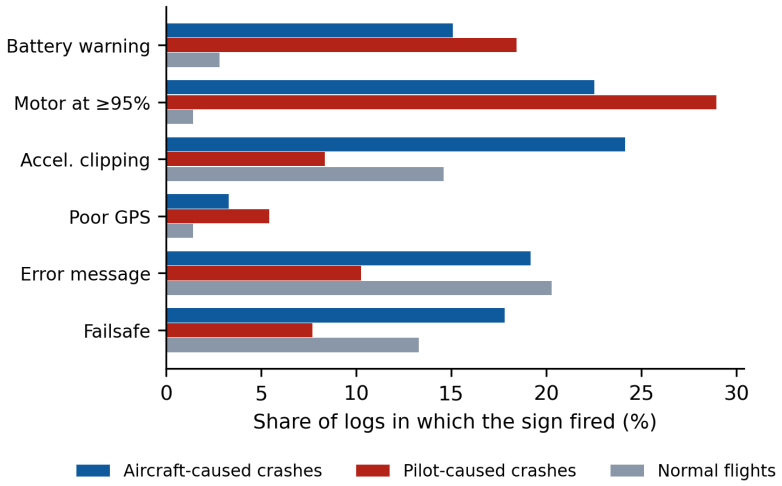


Figure 2: How often each aircraft-fault sign fired in held-out crashes, by uploader label, and in normal flights.

1,000-iteration permutation null distribution on Task B by randomly shuffling crash attribution labels across folds: the null distribution had mean 0.497 (SD 0.090) with an empirical 95% interval of 0.313–0.660. The observed cross-validated AUC of 0.449 falls comfortfully within this chance range (two-sided permutation $p = 0.509$).

To evaluate whether non-linear feature interactions could extract subtle pre-crash signatures, we trained gradient-boosted decision tree ensembles on Task B. Depth-1 decision stumps reached an AUC of 0.560 ± 0.117 at $m = 0\,\text{s}$, falling to 0.413 ± 0.129 at $m = 10\,\text{s}$. Depth-3 boosted trees reached an AUC of 0.595 ± 0.130 at $m = 0\,\text{s}$ (0.604 ± 0.095 at $m = 2\,\text{s}$, 0.621 ± 0.105 at $m = 5\,\text{s}$), falling to 0.553 ± 0.171 at $m = 10\,\text{s}$, 0.579 ± 0.150 at $m = 20\,\text{s}$, and 0.548 ± 0.104 at $m = 30\,\text{s}$. Both boosted ensembles remain within the 95% permutation chance interval (0.313–0.660). Modeling non-linear interactions does not rescue crash attribution; once the impact window is excluded, classification remains near chance. We conclude that crash attribution is not achievable with these features and labels.

5.3. The crash window (Q3a)

In the primary declared sample, crash flights were separated from normal flights with a cross-validated AUC of 0.88 (SD 0.01) at $m = 0\,\text{s}$ ($n = 148$ crashes, 139 normal flights), 0.88 at $m = 2\,\text{s}$ (142 vs. 137), 0.88 at $m = 5\,\text{s}$

(135 vs. 133), 0.87 (SD 0.01) at $m = 10\,\mathrm{s}$ (114 vs. 130), 0.86 at $m = 20\,\mathrm{s}$ (100 vs. 128), declining to 0.86 at $m = 30\,\mathrm{s}$ (89 vs. 124); the declared drop of 0.10 was not observed (observed drop of 0.01). To evaluate whether this apparent stability was an artefact of sample attrition as shorter flights dropped out at larger margins, we evaluated the cross-validation on the fixed constant subset of 213 flights with duration $\geq 40\,\mathrm{s}$ (89 crashes, 124 normal flights; plotted in Fig. 3) across all six margins. On this constant sample, model AUC declined smoothly from 0.895 (SD 0.007) at $m = 0\,\mathrm{s}$ to 0.876 at $m = 2\,\mathrm{s}$, 0.873 at $m = 5\,\mathrm{s}$, 0.868 at $m = 10\,\mathrm{s}$ (a drop of 0.027), 0.862 at $m = 20\,\mathrm{s}$, and 0.864 at $m = 30\,\mathrm{s}$. The absence of a large drop holds on identical flights, confirming that changing sample composition across cut-offs did not mask a collapse.

Leakage was nevertheless present for individual checks. The rate-tracking feature alone fell from an AUC of 0.81 to 0.69 as the crash was excluded, while vibration stayed between 0.74 and 0.77. A fixed tuning check that found 10 of 12 expert-diagnosed tuning faults with the crash included found 1 of 9 once the last 10 s were removed (Section 5.6).

Crucially, the high cross-sectional model separation (AUC 0.88) does not demonstrate unconfounded anticipation of an impending crash: crash and normal flights differ fundamentally in duration (median 54 s against 388 s), firmware era distribution, and operational flight envelope. Table 6 gives the single-feature AUCs. Values below 0.5 indicate features that are lower in crash flights: crash flights reached lower tilt and smaller stick deflections over the whole flight than normal flights, consistent with short, aborted or cautious testing flights. When the exact same 11-feature model was evaluated *within the same drone*—separating each crashed drone's pre-crash window (cut 10 s before impact) from its own earlier flights across 67 multi-flight drones (372 flights: 67 pre-crash flights and 305 earlier flights, distinct from the 372 between-drone control flights)—the pooled Group-CV AUC fell from 0.88 to 0.580 (95% bootstrap CI 0.528–0.633), and the mean within-drone pairwise AUC was 0.605 (95% bootstrap CI 0.516–0.692). Matching on the airframe substantially attenuates the cross-sectional separation from AUC 0.88 to near-chance (AUC 0.58), demonstrating that cross-sectional models are dominated by differences between drone types, flight styles, and durations rather than detecting impending failure, in agreement with the within-drone rank test ($\bar{r} = 0.48$).

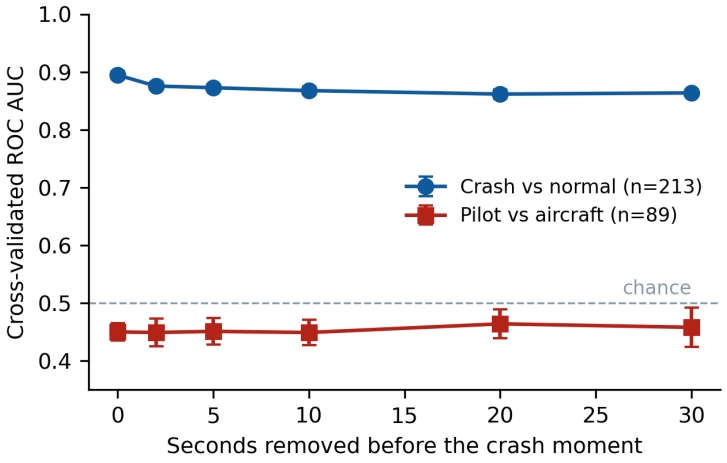


Figure 3: Cross-validated AUC as the end of the analysis window moves away from the crash moment. Bars: standard deviation over five repeats of five-fold cross-validation.

5.4. Within the same drone and between drones (Q3b, Q3c)

Relative to the same drone's earlier flights, the crash flight was neither more vibrating ($\bar{r} = 0.48$, 95% CI 0.38–0.58, $p = 0.67$, 61 drones) nor worse at tracking its commands ($\bar{r} = 0.50$, 0.38–0.63, $p = 0.47$, 46 drones; Fig. 4). Presented as an exclusion bound, the upper 95% confidence bound of 0.58 rules out pre-crash vibration surges within the same drone of the magnitude observed between drones ($\bar{r} = 0.67$). Relative to matched drones without a recorded crash, drones that later crashed vibrated more on their ordinary pre-crash flights ($\bar{r} = 0.67$, 0.57–0.77, $p = 0.0023$, 60 sets; Holm-adjusted $p = 0.016$), while the tracking error did not differ significantly ($\bar{r} = 0.57$, 0.44–0.70, $p = 0.15$, 47 sets). The vibration difference was clearest for aircraft-caused crashes ($\bar{r} = 0.67$, $p = 0.005$, 46 sets) and similar in size but not significant for pilot-caused ones ($\bar{r} = 0.64$, $p = 0.16$, 14 sets).

As a screening signal, the crashed drone was the roughest of its matched set in 33 of 60 sets (55.0%, Wilson 95% CI 42.5–66.9%, against 40.3% expected under the exact Poisson-binomial null; exact one-sided Poisson-binomial $p = 0.0137$). Furthermore, in 38 of 60 sets (63.3%, Wilson 95% CI 50.7–74.4%), the crashed drone exceeded the third quartile (Q3) of its matched controls. Sensitivity differed between measurement sources: 32 of 48 crashed drones (66.7%, Wilson 95% CI 52.5–78.3%) exceeded control Q3 on the older estimator metric, compared to 6 of 12 (50.0%, Wilson 95% CI

25.4–74.6%) on the newer accelerometer metric.

In practical fleet operations, positive predictive value (PPV) depends heavily on the operational flag rate and the fleet base rate. In the archive, 284 distinct vehicles accounted for all 520 real-hardware crash-rated logs out of 38,320 distinct vehicle identifiers, establishing an empirical vehicle crash base rate of 0.741% (284/38,320; or 1.54% if computed as 590 crash logs over 38,320 vehicles). We evaluate screening performance under two distinct operational paradigms:

Empirical matched-set screening (45.1% flag rate): In our matched sets, flagging drones that exceed the upper quartile of their matched controls yields a 63.3% sensitivity (38 of 60). Because sets contain 1 or 2 controls, the weighted null flag rate is 45.1%. At the true vehicle base rate (0.741%), this yields an operational PPV of 1.04% (95% bootstrap CI 0.83–1.25%), an unflagged risk of 0.496%, a relative risk of 2.09× over unflagged airframes, a lift of 1.40× over base rate, and 96.4 inspections per detected crash. At the crash-log base rate (1.54%), the matched-set PPV is 2.14% (relative risk 2.15×, lift 1.39×, 46.7 inspections per crash).

Idealized fleet-wide reference distribution (25% flag rate): If an operator compares airframes against an established fleet-wide baseline and flags the top 25%, the estimated risk in flagged drones is 1.86% versus 0.363% in unflagged drones at the vehicle base rate (relative risk 5.10×, lift 2.50×, 53.9 inspections per crash). At the crash-log base rate (1.54%), the PPV reaches 3.81% (95% bootstrap CI 3.18–4.62%) versus 0.759% in unflagged drones—a relative risk of 5.02×, a lift of 2.47× over base rate (fixing the 2.53× typo from prior drafts), requiring 26.2 inspections per detected crash. Flagging the single top-ranked drone in each set yields a risk of 3.33% in flagged versus 0.93% in unflagged (lift 3.58×, 30.1 inspections per detected crash). In all scenarios, the vibration metric functions as a broad maintenance inspection filter rather than an operational crash warning.

5.5. Robustness of the between-drone result (exploratory)

The vibration result held when controls were restricted to the same airframe as the case ($\bar{r} = 0.69$, $p = 0.008$, 39 sets), when case flights from the crash day were removed ($\bar{r} = 0.76$, $p = 0.003$, 25 sets) and after adjusting for flight duration ($\bar{r} = 0.69$, $p = 0.0005$, 57 sets; Fig. 5). Duration was barely correlated with vibration (Spearman $\rho = 0.05$ and -0.04 for the two measurement sources), although case flights were longer than control flights (median 289 s against 173 s). The sharp sample shrinkage when removing

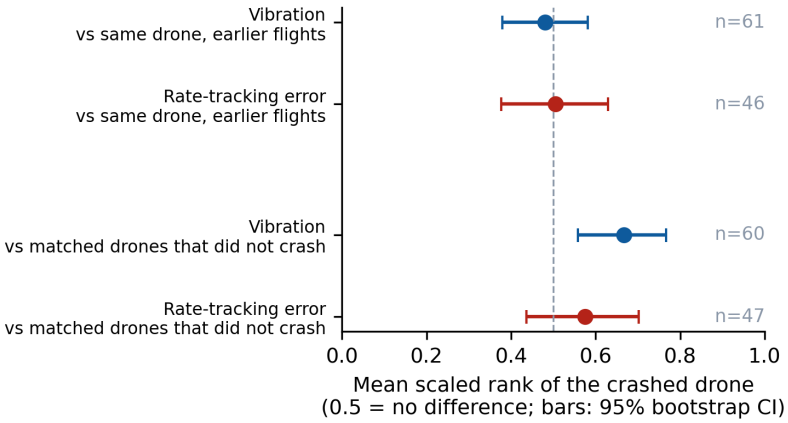


Figure 4: Mean scaled rank of the crashed drone against its own earlier flights (top) and against matched drones without a recorded crash (bottom). 0.5 indicates no difference.

crash-day flights (from 60 to 25 sets) indicates that many pre-crash flights in the public archive occurred during the same flight session or day. The archive index did not record intra-day flight sequence, though parsing raw GPS timestamps in future work could establish whether crash-day flights immediately preceded the crash. This points to potential confounding by development or tuning sessions rather than progressive operational degradation: the "maintenance view" (vibration reflecting mechanical wear over time) competes directly with an "experimental airframe" view (poorly tuned or prototype drones vibrate more and crash during testing). Furthermore, matching was limited to flight controller hardware, PX4 major.minor version and vehicle type; it did not control for propellers, payload, airframe size, or operator identity, so controls from the same operator were not clustered.

The vibration difference was carried by the older estimator metric ($\bar{r} = 0.68$, $p = 0.005$, 48 sets); within the original cohort, the exploratory subset with the newer accelerometer metric showed a smaller, non-significant difference ($\bar{r} = 0.62$, $p = 0.191$, 12 sets; Fig. 5). A prospective replication protocol on newer accelerometer logs selected 26 sets, yielding 19 usable matched sets ($\bar{r} = 0.58$, 95% CI 0.39–0.76, one-sided $p = 0.251$), of which 10 sets were new cases entirely outside the original test ($\bar{r} = 0.50$, 95% CI 0.20–0.75, $p = 0.576$). Statistical power analysis at one-sided $\alpha = 0.025$ demonstrates that detecting $\bar{r} = 0.67$ ($\Delta = 0.17$) with 80% power requires 55 matched sets, while detecting $\bar{r} = 0.58$ ($\Delta = 0.08$) requires 247 sets. Given $N = 19$,

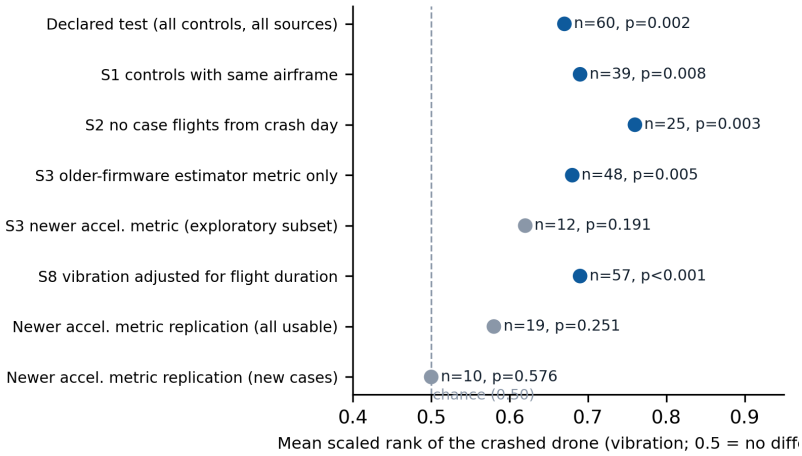


Figure 5: Between-drone vibration result under alternative specifications (exploratory). Blue: $p < 0.025$.

the observed post-hoc power if the true effect were $\bar{r} = 0.67$ was only 0.379 (and 0.118 if $\bar{r} = 0.58$). The newer metric result is therefore severely underpowered and inconclusive rather than negative; a prospective replication on at least 55 modern airframe sets is required.

5.6. Expert diagnoses (Q4)

Of 148 forum threads describing a real crash, the expert attributed it to the aircraft in 94 (64%), to the pilot in 9 (6%), to both in 2, and was unclear or gave no diagnosis in 43 (29%). Tuning or configuration was the cause named most often (26 of 94; 14 of the 47 confirmed by the poster), followed by GPS, compass or estimator problems (18) and motors, ESCs or propellers (16) (Fig. 6). Of 14 threads whose log carried an aircraft crash rating, the expert agreed in 11 and was unclear in 3; none attributed the crash to the pilot. Only 37 of the 73 logs behind aircraft diagnoses with a checkable cause could still be downloaded (38 total downloadable aircraft crash logs). Table 7 shows the fault checks with and without the crash window. With the window ending 10 s before the crash, no check met the declared criterion. The tuning check illustrates the diagnostic leakage described in Section 5.3: across the 9 constant tuning logs with duration ≥ 10 s present in both windows, firing fell from 7 of 9 (77.8%) with the full window to 1 of 9 (11.1%) at $m = 10$ s (6 logs lost firing, 0 gained; exact two-sided McNemar $p = 0.03125$). Across the other four checks (vibration, GPS, motor, battery), firing rates were identical

between full window and $m = 10\,\mathrm{s}$ on paired logs ($p = 1.0$). Data leakage was therefore specific to rate-tracking and tuning fault detection.

To test diagnostic attribution on independently evaluated incidents, we evaluated the rule and models across all 38 downloadable logs behind expert forum threads. Because all 38 incidents were independently diagnosed by autopilot engineers as aircraft faults (no pilot-caused crashes with active log links were found in the forum), this serves as a one-class sensitivity and false-negative check. When evaluated on these real incidents, the fixed rule predicted "aircraft" in 18 (47.4%), "unknown" in 6 (15.8%), and erroneously blamed the "pilot" in 14 (36.8%). On decided cases ($n = 32$), the rule was wrong on 43.8% (14/32). Detailed inspection of flight mode sequences confirmed the failure mechanism: in all 14 misclassified logs, the vehicle was in a pilot-flown mode at the crash moment (`POSCTL` in 7 logs, `STAB` in 4, `ALTCTL` in 3; including log `e9034973`, which transitioned `POSCTL` → `AUTO_4` → `POSCTL` as the pilot attempted manual recovery). Because no hardware failsafe fired before impact, the rule defaulted to blaming the pilot. Similarly, when evaluated out-of-fold on the 30 logs with complete telemetry features, the logistic regression model predicted "aircraft" in 27 (90.0%) and "pilot" in 3 (10.0%), defaulting strongly toward its 72.3% aircraft training prior. This empirical test on independent incidents confirms that neither telemetry rules nor models reliably attribute responsibility.

5.7. Language-model review panels (summary)

Because language-model assistants are increasingly proposed for log review, we conducted a controlled exploratory evaluation of local models (3–4B parameters) configured as single agents and multi-agent review panels on 60 author-generated fault cases and worst-case adversarial prompts. As detailed in Appendix B, the fixed rule reached 75% right and 0% wrong (referring 25% to humans), whereas local language models produced substantial error rates (10–63% wrong), suffered from mode collapse or abstention on active causes, and proved vulnerable to consensus manipulation by a single compromised reviewer (flipping 45–56 of 60 cases). In contrast, deterministic numeric telemetry extraction prevented direct prompt injection by construction.

5.8. Analyses added during revision (exploratory)

To preserve the boundary between pre-declared confirmatory hypotheses and exploratory post-hoc investigations [17], Table 8 documents the analyses

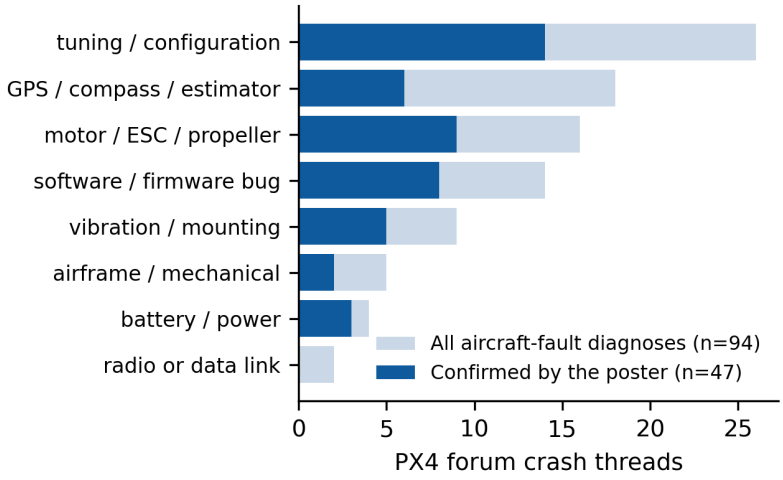


Figure 6: Causes named by experts in PX4 forum threads that attributed a crash to the aircraft.

added during peer review, their motivation, key results, and the git commit hashes where their declarations or implementations were introduced. None of these exploratory additions altered the conclusions of the pre-declared tests; rather, they provided empirical bounds on confounding, evaluated alternative non-linear architectures, and quantified operational screening performance.

Table 3: Analysis samples and evaluation cohorts.

Sample	Crashes	Comparison	Question
Settings	100 (57 with default records)	477 normal (113 with default records)	Q1
Development crashes	93 analysed	–	Q2 rule
Held-out crashes	112 analysed	143 normal (full logs)	Q2 replication
Leak-curve set	148 at $m = 0\,\mathrm{s}$ (89 at $m = 30\,\mathrm{s}$)	139 normal (124 at $m = 30\,\mathrm{s}$)	Q2 model, Q3a
Constant-subset leak curve	89 crashes ($\geq 40\,\mathrm{s}$ duration)	124 normal ($n = 213$ constant across margins)	Q2 model, Q3a
Within-drone rank test	101 selected (61 and 46 analysed)	475 earlier flights of the same drones	Q3b
Within-drone predictive model	67 multi-flight drones (67 pre-crash)	305 earlier flights of same drones ($n = 372$)	Q3b
Between-drone matched sets	66 matched sets (60 and 47 analysed)	124 matched drones, 372 flights	Q3c
Expert forum cases	148 real crash threads	38 logs downloadable (all 38 aircraft-fault)	Q4

Table 4: Features computed on each analysis window.

Feature	Definition
Rate-tracking error	RMS of commanded minus measured body rate, roll and pitch (rad s^{-1}; rotary wing)
Tilt (p95)	95th percentile of tilt angle (deg)
Vibration (p95)	95th percentile of the accelerometer vibration metric, or of the estimator vibration estimate on older firmware
Clipping rate	accelerometer clipping events per second
Motor saturation	share of time with any motor output at 95% or more (rotary wing)
Battery minimum	lowest reported remaining charge (%)
Voltage sag	(start voltage − minimum voltage) / start voltage
Poor GPS	share of time with horizontal accuracy above 5 m or fewer than 6 satellites
Failsafe	1 if a failsafe was active at any time
Manual share	share of time in a pilot-flown mode (manual, altitude, position, acro, stabilised)
Stick deflection (p95)	95th percentile of the larger roll or pitch stick deflection

Table 5: Confirmatory tests, their declared pass marks and outcomes. Holm adjustment over the seven p-valued tests.

Test (declared criterion)	Result	Holm p	Outcome
Protections disabled, crash vs normal	51% vs 46%, $p = 0.63$	1.00	No difference
Attribution rule, dev. (BA ≥ 0.70)	BA 0.57, 35% wrong	–	Failed
Attribution rule, held-out	BA 0.47, 47% wrong	–	Failed
Clipping, aircraft vs pilot ($p < 0.025$)	24% vs 8%, $p = 0.045$	0.27	Not supported
Manual mode, pilot vs aircraft ($p < 0.025$)	18% vs 10%, $p = 0.165$	0.73	Not supported
AUC drop from 0 to 10 s (≥ 0.10)	0.88 to 0.87	–	Not supported
Within-drone vibration ($p < 0.025$)	$\bar{r} = 0.48$, $p = 0.67$	1.00	Not supported
Within-drone tracking ($p < 0.025$)	$\bar{r} = 0.50$, $p = 0.47$	1.00	Not supported
Between-drone vibration ($p < 0.025$)	$\bar{r} = 0.67$, $p = 0.0023$	0.016	Supported on older metric only; replication inconclusive
Between-drone tracking ($p < 0.025$)	$\bar{r} = 0.57$, $p = 0.15$	0.73	Not supported
Expert-fault checks, $m =$ 10 s ($\geq 50\%$, $\leq 10\%$)	best: vibration 3/5 vs 11%	–	None met

Table 6: Single-feature AUC for separating crash from normal flights at three cut-offs (no fitting; 0.5 is chance; values below 0.5 mean lower in crash flights).

Feature	$m = 0\,\mathrm{s}$	$m = 10\,\mathrm{s}$	$m = 30\,\mathrm{s}$
Rate-tracking error	0.81	0.72	0.69
Vibration (p95)	0.74	0.72	0.77
Manual share	0.67	0.68	0.69
Motor saturation	0.67	0.66	0.65
Clipping rate	0.59	0.54	0.55
Voltage sag	0.55	0.53	0.48
Poor GPS	0.53	0.55	0.54
Failsafe	0.50	0.47	0.47
Battery minimum	0.42	0.40	0.38
Tilt (p95)	0.41	0.36	0.33
Stick deflection (p95)	0.32	0.33	0.35

Table 7: Fault checks on logs whose expert named the matching fault, with the full window and with the last 10 s before the crash removed. Declared criterion: at least 50% of diagnosed logs and at most 10% of normal flights.

Expert-named fault	Full window: diagnosed	Full window: normal	$m = 10\,\mathrm{s}$: diagnosed	$m = 10\,\mathrm{s}$: normal
Tuning or configuration	10/12 (83%)	18/142 (13%)	1/9 (11%)	7/130 (5%)
Vibration or mounting	3/5 (60%)	7/48 (15%)	3/5 (60%)	4/38 (11%)
GPS, compass or estimator	1/5 (20%)	10/143 (7%)	1/4 (25%)	5/130 (4%)
Motor, ESC or propeller	1/9 (11%)	8/143 (6%)	0/4	5/130 (4%)
Battery or power	0/4	4/143 (3%)	0/2	1/130 (1%)

Table 8: Exploratory analyses added during peer review, their motivation, key results, and git commit hashes.

Analysis	Motivation	Result	Commit
1,000-permutation null (Task B)	Calibrate below-chance CV AUC	Null mean 0.497 [0.313, 0.660]; two-sided $p = 0.509$	`bad06ff`
Boosted decision trees (Task B)	Test non-linear feature interactions	Stumps $0.560 \to 0.413$; Depth-3 $0.595 \to 0.553$ at $m = 10\,\mathrm{s}$	`bad06ff`
Constant-subset leak curve	Exclude sample attrition ($n = 213$)	AUC declines smoothly $0.895 \to 0.868$ ($m = 10\,\mathrm{s}$)	`bbdd042`
Paired diagnostic leakage test	Test leakage on constant logs ($n = 9$)	Tuning check $7/9 \to 1/9$, exact McNemar $p = 0.03125$	`bbdd042`
Within-drone predictive model	Account for airframe confounding	Group-CV AUC 0.580; mean within-drone AUC 0.605	`bad06ff`
Poisson-binomial set null	Account for mixed set sizes ($n = 60$)	Exact null rate 40.28%; observed 55.0% ($p = 0.0137$)	`bad06ff`
Screening PPV and CIs	Quantify operational fleet value	Matched PPV 1.04% (lift 1.40×); Fleet-ref PPV 3.81% (lift 2.47×)	`bad06ff`
Sensitivity by vibration source	Compare older vs. newer metrics	Older: 32/48 (66.7%); newer: 6/12 (50.0%)	`bad06ff`
38 expert crash logs attribution	Evaluate against independent causes	One-class test: rule wrong on 43.8%; model 90% drone	`bad06ff`

6. Design implications: an assurance tool built on the findings

The authors develop AeroFleet 2.0, an open-source, ground-side assurance tool for PX4 and ArduPilot fleets. It reads the files an operator already has (parameter files, mission files and flight logs), contacts no aircraft and calls no language model. This section reports design consequences and engineering checks; the paper's empirical evidence is presented in Sections 3–5. Table 9 links each result above to a design decision and to the check that was made of it. AeroFleet 2.0's checks are primarily engineering validation tests, not evidence of improved operational safety; demonstrating safety benefits in field operations requires prospective fleet evaluation and user studies, which remain future work (Section 7). The injection, search, and replay validation checks are detailed in Appendix A.

Evidence, not verdicts. A flight review reads an ArduPilot or PX4 log and the approved mission file and lists, in time order, mode changes (with a pilot taking over from an automatic mode flagged), settings changed in flight, battery and voltage warnings, poor GPS and deviations from the plan. It reports the quality of that evidence (sample rate, satellite count and dilution of precision, height reference, clock source) so that a reader can judge how far to trust it, and leaves the conclusion to a person. Given the attribution results, a tool that issued a pilot-or-aircraft verdict would claim more than the log supports.

Ranking drones, not predicting flights. Each reviewed flight's vibration is stored with its measurement source; drones are ranked against others measured the same way, and those above their group's upper quartile are flagged for mechanical inspection. In the archive, this ranking identified the drone that later crashed as the roughest in 55.0% of matched sets (against 40.3% expected under the exact Poisson-binomial null, $p = 0.014$; lift 3.6×), and 63.3% exceeded the control third quartile. In operational terms, this translates to a PPV of 1.04% (95% bootstrap CI 0.83–1.25%, lift 1.40×, 96.4 inspections per crash) under empirical matched-set screening at the 0.741% vehicle base rate, and a PPV of 3.81% (95% CI 3.18–4.62%, lift 2.47×, 26.2 inspections per crash) under an idealized fleet-wide 25% flag rate at the 1.54% log base rate (1.86% PPV at the vehicle base rate). The flag is therefore a maintenance inspection prompt, not a crash warning.

Table 9: From findings to design decisions in the AeroFleet 2.0 assurance tool, and how each part was checked.

Finding	Design decision	Check
Logs did not attribute crashes (Q2); checks fired on the crash (Q4)	A flight review lists the log's events in time order and states the quality of its evidence; it names no cause	Unit and API tests
Roughness marked drones, not flights (Q3b, Q3c)	Drones ranked by vibration within a fleet, measured the same way; the upper quartile flagged for inspection	Unit tests; no prospective evaluation yet
Useful evidence is longitudinal	Every upload, approval and review appended to a hash-chained, Ed25519-signed record with SHA-256 fingerprints of the files received	Tamper tests: edited, deleted, reordered and re-signed records detected
A plan proves nothing if something else was flown	Planned-versus-flown check: deviations beyond a fleet's corridor and height tolerance	Known deviations injected into 40 real flights (Appendix A.1)
Expert diagnoses as precedents (Q4)	Description search over 161 expert-labelled threads; checklist in order of expert frequency	Leave-one-out test on 96 cases (Appendix A.2)
Separation and deviations are three-dimensional	Tabletop 3D replay of a reviewed flight and rehearsal of planned missions, on screen or in a headset	Functional checks only; no user study (Appendix A.3)

A record that can be checked. Because the useful evidence accumulates per aircraft, results are appended to a hash-chained record in which each entry is signed and carries the SHA-256 of the exact files it came from; each new head is also written to a separate file intended to be copied elsewhere. A standalone verifier checks an exported record; in tests it detected an edited field, a deleted entry, a reordering, and a chain rebuilt with a new key (the last only against a pinned key fingerprint or the separately kept heads, which is the stated limit of the design).

7. Discussion

Attribution. A declared rule, its held-out replication and a cross-validated model all found no usable separation between pilot- and aircraft-caused crashes with these labels and features. While commercial and legal disputes often seek a binary verdict, aviation safety science demonstrates that accidents are multi-causal systemic events involving organizational, environmental and human factors [7]. Uploader labels are noisy, and the sharp divergence between uploader pilot blame (32%) and expert forum diagnoses (6%) points to politeness bias on public forums or a narrower definition of pilot fault among autopilot engineers. A log records the combined physical outcome of pilot inputs, environmental disturbances and vehicle response; disentangling responsibility from telemetry alone is not supported by the data. Tools that issue automatic responsibility verdicts should be evaluated against independent investigations before adoption.

Leakage and confounding. While cross-sectional models showed high apparent separation (AUC 0.88), this was dominated by between-drone confounds; within-drone evaluation demonstrated attenuation to near-chance (AUC 0.58), confirming that longitudinal baseline comparison is essential. Diagnostic checks also exhibited leakage: specifically, tuning fault detection fell from 77.8% to 11.1% sensitivity ($p = 0.031$) when the final 10 s were removed. Diagnostic evaluations must remove the crash window and compare against an aircraft's own operating baseline before claiming pre-crash anticipation.

Drone-level risk. The positive result concerns airframes, not flights. Drones that later crashed vibrated more than comparable drones on ordinary flights, consistent with a maintenance view in which mechanical degradation precedes catastrophic failure [24]. However, two important qualifications apply.

First, removing crash-day flights reduced usable sets from 60 to 25, indicating that many flights occurred during the same session; the "maintenance degradation" hypothesis therefore competes with an "experimental airframe" hypothesis, where poorly tuned or prototype drones vibrate heavily and crash during initial testing. Second, the effect was supported on the older estimator metric only; a declared replication on the newer accelerometer metric was inconclusive ($p = 0.25$). Finally, given the low baseline crash rate in active fleets ($\sim 1.5\%$), flagging upper-quartile vibration yields an operational PPV of 3.8% (95% CI 3.2–4.6%; 3.3% for top-rank flagging). The metric provides a maintenance prioritization signal, not an operational warning.

From findings to a tool. Section 6 turns these results into design decisions for AeroFleet 2.0, an operator's ground-side assurance tool. The common thread is that the tool shows evidence and ranks drones, and does not name a cause.

Language models. On this task small local models added errors and voting vulnerabilities rather than accuracy. While deterministic numeric parsing prevented direct prompt injection by construction, multi-agent voting consensus was easily manipulated by a single compromised reviewer. Larger models and blind evaluation cases remain to be tested.

Relation to prior datasets. Benchmarks built from injected faults [8, 9] answer whether a known fault can be detected; anomaly datasets from real logs [10] answer whether unusual segments can be found. Our results concern a different question, attribution of a real crash, and suggest that performance on the first two does not transfer to it: signals that mark a fault also mark the crash it causes and the pilot's reaction to it. Datasets that pair real crashes with independently verified causes would be needed to make progress on attribution, and our expert-diagnosis labels are a small first step.

Records and regulation. Record-keeping rules ask operators to keep logs, maintenance records and occurrence reports [1, 2]. On our evidence, such records are most useful when they are kept per aircraft over time, so that an airframe's condition can be compared with its own history and with comparable airframes, rather than examined only after an accident.

Practical guidance. For operators and investigators we suggest: (i) treat automatic pilot-or-aircraft verdicts from logs with caution; (ii) use logs to assemble a time-ordered record of events for a human investigator; (iii) track

vibration per drone across flights and inspect drones that are rough relative to comparable airframes; (iv) when evaluating any log-based diagnostic, remove the crash window and compare against the same drone's history.

Future work. Two studies have been declared in the repository before any data were collected: a prospective test in partner fleets, in which the monthly vibration ranking of AeroFleet 2.0 is written to the signed record before outcomes occur and crashes, unplanned repairs and defects found at inspection are compared between flagged and unflagged drones over six months; and a within-subjects user study comparing the three-dimensional views with a two-dimensional map on tasks with known answers. Stronger replication of the between-drone vibration effect on modern autopilots, evaluating operational safety impact across active fleets, testing sequence models or frequency-domain IMU features for transient precursors that whole-window summaries miss, and evaluating a larger set of crashes with causes verified by independent investigators would further strengthen the evidence.

8. Threats to validity

Archive self-selection. The public Flight Review archive reflects voluntary uploads from hobbyists, academic researchers, and commercial operators, introducing self-selection towards unusual flights, developer tuning sessions, and unexplained incidents. Crashed airframes and normal comparison flights differ substantially in flight duration (median 54 s vs. 388 s), firmware era, and operational envelope, which motivated our within-drone and matched-pair designs.

Labels. The uploader-assigned responsibility labels are not independently verified ground truth; they reflect the uploaders' own assessments, and normal flights are self-rated. The expert-diagnosed sample confirmed agreement on aircraft-rated logs (11 of 14) but did not evaluate pilot-rated logs, and the gap between expert (6%) and uploader (32%) pilot blame rates suggests reporting biases. Controls without a crash-rated log may have crashed without a rating, biasing the between-drone test conservatively. Forum diagnoses were coded with good inter-rater reliability ($\kappa \geq 0.73$), but were limited to public discussions.

Confounding and design. Matching on vehicle type, controller hardware and PX4 version leaves open differences in propellers, payload, flight style or operator identity, leaving potential unmodeled clustering. Results may not transfer to commercial ArduPilot fleets or to proprietary autopilots, even though the tool is designed to support both ArduPilot and PX4 logs.

Measurement. The crash moment is estimated; the within-drone result did not depend on the threshold. The between-drone effect was shown mainly with the older vibration estimate and was not confirmed on the newer metric in a declared replication with 19 usable sets; this main positive result needs stronger replication on newer firmware.

Protocols and registration. Protocols were declared and committed to public version control in advance of analysis, but git repositories are self-hosted and lack third-party timestamped registries (e.g. OSF). One amendment (lifting the within-drone firmware filter) was data-dependent after observing sample starvation ($n = 12$).

Features. Summary features over the whole window (percentiles, shares of time, RMS) were used; sequence models or frequency-domain IMU features might capture short precursors that summaries miss. This was not tested.

Attrition. Only 37 of the 73 logs behind expert aircraft diagnoses with a checkable cause could still be downloaded, so several diagnostic comparisons in Table 7 have very small sample sizes ($n = 2$ to 9); its results are indicative rather than definitive.

Power. Some comparisons are small (14 pilot-caused matched sets, 12 sets with the newer vibration metric, five vibration-diagnosed logs). Null results there are not evidence of no effect.

Language models. The AI comparison is preliminary: the experiments use 60 author-generated cases and relatively small local models (3–4B parameters). The fixed rule's performance may be optimistic on cases created by the same authors who designed the rule (0 wrong answers is close to a training score); a blind set of cases written by someone who has not seen the rule is in preparation. Larger models or alternative prompt architectures were not tested.

The tool. AeroFleet 2.0's checks are primarily engineering tests on archive data and synthetic fault injections, not evidence of improved operational safety. Plans in the injection test were built from the flights themselves; the description search was tested on forum titles, which are shorter and more varied than an operator's report, and its symptom lists were written by the authors; prospective fleet evaluation of the vibration ranking and user studies of the three-dimensional views remain future work (Sections 6 and 7).

9. Conclusion

Across 590 crash-rated logs from an archive index of 479,547 flights, flight logs did not distinguish pilot-caused from aircraft-caused crashes above chance, configuration protections were disabled at similar rates, and the flight that crashed showed no detectable difference in vibration or rate-tracking error relative to the same drone's prior flights (95% CI upper bounds 0.58 and 0.63). Diagnostic checks also suffered from crash-impact data leakage: tuning fault detection fell from 77.8% to 11.1% sensitivity ($p = 0.031$) once the final 10 s were removed. What the logs did reveal, on older estimator metrics, was that drones that later crashed exhibited higher baseline vibration on ordinary flights than matched peers, though replication on newer accelerometer metrics was inconclusive ($p = 0.25$) and operational screening yields an operational positive predictive value of 1.04% under empirical matched-set screening (3.8% under a theoretical fleet-wide upper-quartile reference). Flight log analytics for drone fleets should prioritize longitudinal airframe screening and structured evidence review, not automated verdicts. AeroFleet 2.0 implements this approach as an open-source tool, presenting verifiable evidence while leaving diagnostic decisions to human operators.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The authors develop the open-source AeroFleet 2.0 tool described in Section 6; it is not commercial.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used an AI assistant (Claude, Anthropic) to write analysis code, draft and revise text, and assist the recording of forum diagnoses. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Data availability

Selection scripts, analysis code, the declared hypotheses with their commit history, derived features, expert labels (without forum text), and the source code of the AeroFleet 2.0 tool are available at `https://github.com/AdityaPathare46/aerofleet`; a DOI-archived release will be provided on acceptance. Raw logs are public on PX4 Flight Review and are referenced by identifier.

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Appendix A. Assurance tool implementation and validation checks

Appendix A.1. Planned versus flown trajectory checks

The check aligns the flown track with the plan's legs in flying order over the whole flight and reports stretches beyond the fleet's sideways and height tolerances (defaults 30 m and 15 m for at least 3 s). To test it, plans were built from 40 real multirotor flights in the archive (a waypoint every 10 s of track), so that the unmodified flight conformed by construction, and a known deviation was injected for 30 s. Sideways shifts of 35 m or more were found in 90–98% of flights and upward shifts of 20 m or more in 92–98%, while shifts below the tolerances were mostly not reported, as specified. The first version of the check, which matched each position to the nearest leg anywhere, raised false alarms on 16 of 40 unmodified flights and detected only half of the 35 m shifts; it was replaced. The current version raised false alarms on 9 of 40 (22%, 95% CI 12–38%), all on fast flights cutting the corners of long legs

built from the track; with a waypoint every 5 s (a post-hoc check) there were none. A declared check on the archive logs that record the mission they flew found only six distinct missions; the check reported no deviation while any of them flew in mission mode, and the two deviations it did report were stretches in which the vehicle had left mission mode (return-to-launch, or the pilot taking over). Six missions are too few for a rate (95% upper bound 39%), so the false-alarm rate on real plans remains unmeasured and a reported deviation is a reason to look at the replay rather than a finding.

Appendix A.2. Searching past community diagnoses

An operator can describe a problem in words. The description is matched (BM25, no model) against the expert-labelled forum threads, represented by their titles and up to 30 descriptive terms from the opening post, and against a short list of symptom words for each cause. After a first release we declared and ran a test, holding out each of the 96 threads in which the expert blamed the aircraft and named a cause and querying with the rest. The cause the words matched best was the expert's cause in 19 of 96 cases with the title alone (20%) and 21 of 96 with the title and opening-post terms (22%, 95% CI 15–31%), below the 26 of 96 (27%) of always answering the most frequent cause; among the first three it was 54% against 62%. The declared criterion (10 points above the base rate) was not met, as it was not with the first recording (20% against 32%). A written description, like the log, did not reveal the cause. The tool therefore orders its checklist by how often experts found each cause, marks the causes the description mentions, and offers the most similar threads only as reading, with the measured accuracy shown beside the answer.

Appendix A.3. Three-dimensional replay and mission rehearsal

Deviations, heights and separation between drones are spatial, and a two-dimensional plot hides height. The tool therefore draws flights on a virtual tabletop, a world in miniature [32], that can be viewed in a browser, in immersive virtual reality or, on a passthrough headset, in mixed reality on a real desk (WebXR).

A *flight replay* shows a reviewed flight: the approved plan dashed, the flown track green within tolerance and red during a deviation, the log's events pinned where they happened, and a playhead that can be scrubbed or played at ten times real speed.

A *mission rehearsal* shows planned missions before anyone flies: the deconfliction check turns each mission file into a position-over-time trajectory on a common clock (straight legs at stated climb, cruise and descent speeds), finds every pair closer than the separation distance and resolves conflicts by a launch delay or, where allowed, a different cruise height. The rehearsal flies all missions on that clock, as uploaded and as rescheduled, with each loss of separation drawn where and when it occurs (Figure A.7). It is a kinematic rehearsal of the plan, not a simulation of the aircraft: there is no wind, no vehicle dynamics and no sensor error, so it shows what the plan implies if it is flown as written, which is the assumption the planned-versus-flown check then tests after the flight.

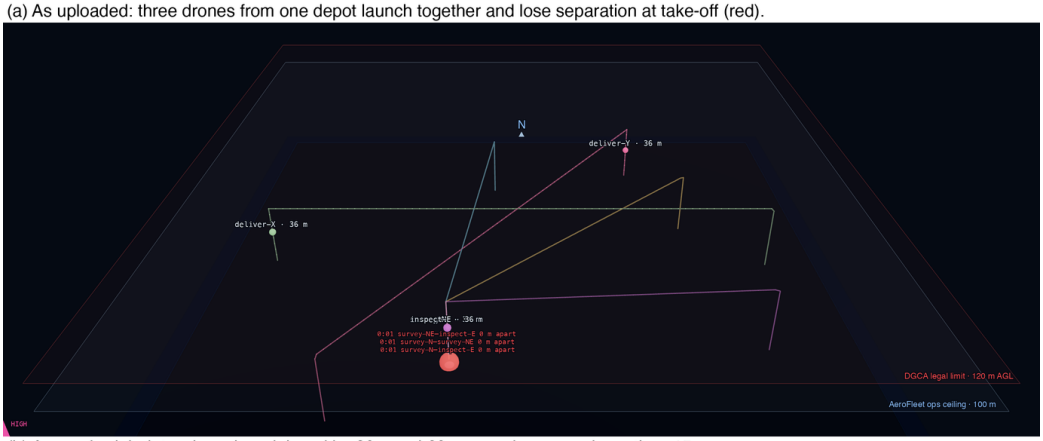
The replay and rehearsal have been checked functionally on desktop; a paired standalone headset application compiles but has not been run on a headset, and no user study has been carried out, so we make no claim that the three-dimensional view improves decisions.

Appendix B. Exploratory evaluation of language-model review panels

Because language-model assistants are increasingly proposed for log review, we conducted a preliminary comparison on 60 author-generated fault cases using four relatively small local models (3–4B parameters). The fixed rule was right in 45 of 60 generated cases (75%, 95% Wilson interval 63–84%) and wrong in none (0%, 95% interval 0–6%); it referred the other 15 to a human operator. "Right" includes leaving the 10 no-cause cases open, so the rule named the true cause in 35 of the 50 cases that had one. As discussed in Section 8, the rule's performance may be optimistic because the cases were created by the authors who designed the rule; nonetheless, no model's panel matched it (Table B.10): wrong answers ranged from 6 to 38 of 60 (10–63%).

Reporting coverage (share of cases decided) and accuracy-when-answering separately clarifies the failure mechanisms of individual models. The single-agent configurations of `gemma3:4b` and `phi4-mini` abstained on all 50 cases with an active cause (0% active coverage); their 10 "right" scores stemmed entirely from correctly leaving the 10 no-cause cases open. `qwen2.5:3b` answered 53 of 60 cases (88% coverage) but achieved only 17.0% accuracy when answering (9 right, 44 wrong), which is indistinguishable from random guessing across the six cause classes (16.7%). `qwen3:4b` suffered complete mode collapse on decided cases: across the 41 cases it answered (the remaining 19

(a) As uploaded: three drones from one depot launch together and lose separation at take-off (red).



(b) As rescheduled: two launches delayed by 30 s and 60 s; no pair comes closer than 15 m.

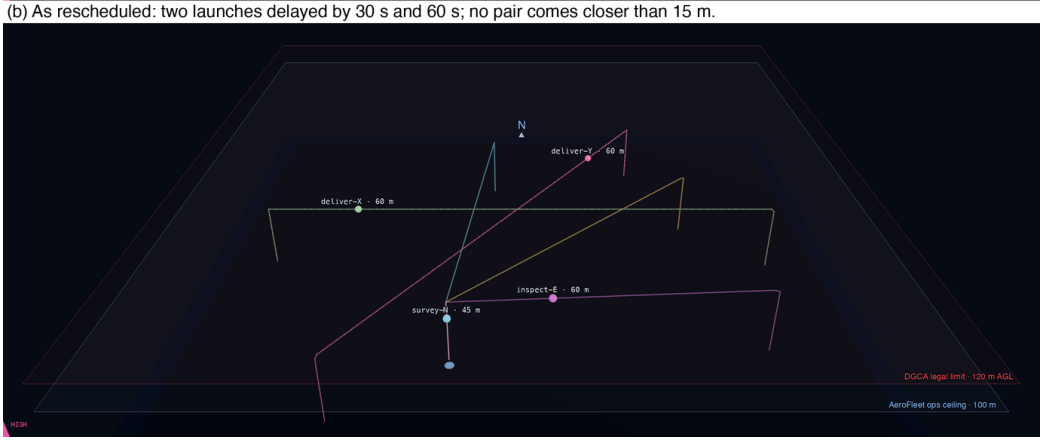


Figure A.7: Mission rehearsal in the tool's tabletop view (desktop rendering; the same scene opens in a WebXR headset). Five planned missions over a 2 km table, heights exaggerated four times, each mission in its own colour. (a) As uploaded, 12 s after launch: three missions leaving one depot together occupy the same point (overlapping labels) and the red sphere marks their loss of separation. (b) The schedule produced by the deconfliction check, 45 s after the first launch: two launches are delayed and no pair comes within 15 m. Straight legs at planned speeds, no wind.

failed JSON schema parsing), it attributed all 41 to `PLAN_OR_ROUTE_ISSUE`, yielding 0 right, 41 wrong (0% accuracy when answering).

Multi-agent panel deliberation increased coverage for `gemma3:4b` and `phi4-mini` to 65%, but accuracy when answering remained low for `gemma3` (7.7%, 3 of 39) and modest for `phi4` (51.3%, 20 of 39). `qwen2.5:3b` achieved 62.5% accuracy when answering in the panel (10 right, 6 wrong), but deferred on 73% of cases (16 answered). `qwen3:4b` reached 30.9% accuracy when answering (17 of 55).

In the adversarial robustness test, we conducted a worst-case sensitivity sweep where a single compromised reviewer systematically advocated for a designated incorrect cause across both voting rounds. This adversary successfully flipped the panel's consensus in 45 to 56 of 60 cases (Table B.10). This highlights that multi-agent consensus voting without cryptographic proofs or fact-citation enforcement remains vulnerable to rogue reviewers. In contrast, 40 hostile prompt-injection strings planted in uploader-controlled fields never reached any model prompt because the pipeline deterministically extracts numeric telemetry into code, preventing direct prompt injection by construction.

Table B.10: Language-model review panels on 60 generated cases: counts of right (R), wrong (W), and deferred (Def) cases, with coverage (Cov., share of cases decided) and accuracy when answering (Acc.). "Right" includes correctly leaving a no-cause case open. Single-agent gemma3 and phi4 answered only the 10 no-cause cases (0% coverage on active causes). Adversary flips show consensus changes caused by a single compromised reviewer in a worst-case sweep.

	Single agent			Review panel			Compromised
Method	R / W / Def	Cov.	Acc.	R / W / Def	Cov.	Acc.	Flips / 60
Fixed rule (no model)	45 / 0 / 15	75%	100%	–	–	–	–
`gemma3:4b`	10 / 0 / 50	17%†	100%†	3 / 36 / 21	65%	7.7%	49
`phi4-mini`	10 / 0 / 50	17%†	100%†	20 / 19 / 21	65%	51.3%	56
`qwen2.5:3b`	9 / 44 / 7	88%	17.0%	10 / 6 / 44	27%	62.5%	54
`qwen3:4b`	0 / 41 / 19	68%	0.0%	17 / 38 / 5	92%	30.9%	45

†Answered only on the 10 no-cause cases where abstention coincided with the ground truth; 0% coverage on active causes.

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